

Car Price Prediction – Examination Notes

Goal of Assignment

- Show ability to use Python + Jupyter Notebook for ML.
- Not about perfect accuracy, but reasoning and explanation.
- Dataset: Car Price Prediction (features: year, mileage, fuel, seller_type, etc.).
- Target: Selling Price.

Workflow

1. Data Loading & Inspection – load car_prediction_data.csv, check shape, missing values.
2. Exploratory Data Analysis (EDA) – histograms of price, scatterplots (mileage vs price), correlation heatmap.
3. Preprocessing – handle missing values, encode categoricals (fuel, seller_type, transmission), scale numerics.
4. Baseline Model – Linear Regression (simple, explainable).
5. Optimization – RandomForest Regressor + GridSearchCV for tuning.
6. Evaluation – RMSE, R^2 , predicted vs actual plot, feature importances.
7. Explanation – Why each step was chosen, not just how.

Baseline Model (Linear Regression)

- Simple and interpretable.
- Provides a benchmark RMSE and R^2 .
- Explained as first attempt without tuning.

Optimization (Random Forest)

- Captures nonlinear relationships better than linear regression.
- Hyperparameter tuning: n_estimators and max_depth.
- Compare before/after with graphs and metrics.
- Show feature importances.

Presentation Structure

1. Introduce dataset and problem (predict car price).
2. Show EDA findings with graphs.
3. Present baseline model results.
4. Present optimization and improvements.
5. Conclude: ML gives better than guessing; explain lessons learned.

Key Reminders

- Always explain WHY you took a step (e.g., why RandomForest).
- Use visuals: histogram of prices, feature importance plot, prediction vs actual scatter.
- Quality is judged on reasoning + documentation, not just high accuracy.
- Final deliverables: GitHub repo (.ipynb) + Azomo submission + oral presentation.